

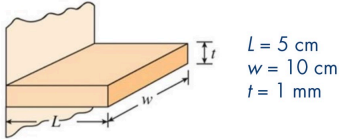
Example

A single straight brass fin with rectangular cross-section is 5 cm long.

The temperature at the base of the fin is 60°C and the surrounding ambient air is 30°C . The brass conducts heat at 110 W/mK and the convective heat transfer coefficient is $10 \text{ W/m}^2\text{K}$.

Treating the fin with an **adiabatic tip** of corrected length, determine:

- The rate of heat transfer from the fin.
- Thermal resistance of the fin.
- Efficiency of the single fin
- Effectiveness of the single fin



$$\begin{aligned} L &= 5 \text{ cm} \\ w &= 10 \text{ cm} \\ t &= 1 \text{ mm} \end{aligned}$$

$$h = 10 \text{ W/m}^2\text{K}$$

$$k = 110 \text{ W/mK}$$

$$\begin{aligned} A_c &= wt = (10 \times 10^{-2} \text{ m})(1 \times 10^{-3} \text{ m}) \\ &= 10^{-4} \text{ m}^2 \end{aligned}$$

$$P = 2(w + t) = 0.202 \text{ m}$$

$$\begin{aligned} \frac{A_c}{P} &= \frac{wt}{2(w+t)} \cdot \frac{(1/w)}{(1/w)} \\ &= \frac{t}{2(1+t/w)} = \frac{t}{2} \end{aligned}$$

$$\begin{aligned} L_c &= L + \frac{A_c}{P} \\ &= L + \frac{t}{2} \\ &= 0.05 + \frac{0.001}{2} \text{ m} = 0.0505 \text{ m} \end{aligned}$$

$$\begin{aligned} \underline{a.} \quad \dot{Q}_{\text{adiabatic}} &= \sqrt{hPkA_c} (T_b - T_\infty) \tanh(mL_c) \quad \text{where} \quad m = \sqrt{\frac{hP}{kA_c}} \\ &= 4.47 \text{ [W]} \tanh(mL_c) \\ &\quad \text{prev. example} \\ &= 4.47 \text{ [W]} \tanh(1355 \text{ m}^{-1} \cdot 0.0505 \text{ m}) \\ &= \boxed{2.66 \text{ [W]}} \end{aligned}$$

$$\underline{b.} \quad R_{\text{fin}} = \frac{T_b - T_\infty}{\dot{Q}_{\text{adiabatic}}} = \frac{60^\circ\text{C} - 30^\circ\text{C}}{2.66 \text{ [W]}} = \boxed{11.288 \text{ [K/W]}}$$

$$\underline{c.} \quad \eta_{\text{fin}} = \frac{\dot{Q}}{hA_{\text{fin}}(T_b - T_\infty)} = \frac{\dot{Q}}{h(P L_c + A_c)(T_b - T_\infty)} = \frac{2.66 \text{ [W]}}{10 \text{ [W/m}^2\text{K]} \times 0.202 \text{ m} \times 0.0505 \text{ m} \times (60^\circ - 30^\circ\text{C})} = \boxed{0.87}$$

Agitjik

$$\underline{d.} \quad \varepsilon_{\text{fin}} = \frac{\dot{Q}}{hA_b(T_b - T_\infty)} = \frac{\dot{Q}}{10 \text{ [W/m}^2\text{K]} \cdot 10^{-4} \text{ m}^2 \cdot (60^\circ - 30^\circ\text{C})} = \boxed{88.51} \gg 1$$

Very effective